

OBJECTIVE

Results-driven **Embedded Engineer with 5 years of experience in C, C++, Linux, and Qt**, aiming to leverage strong programming and system design skills to develop robust and scalable embedded applications. Seeking an opportunity to contribute to cutting-edge technologies while continuously enhancing technical expertise in real-time and safety-critical systems.

EDUCATION

- **Bachelor of Technology in Electronics and Computer Engineering** 2015-18
YSR Engineering College of Yogi Vemana University, Proddatur
- **Diploma in Electronics and Computer Engineering** 2012-15
GMR Polytechnic Srisailam, Srisailam
- **Class X** 2011-12
Wisdom High School, Sundipenta

TECHNICAL SKILLS

Languages: C/C++, Embedded C, Socket Programming
Frameworks: Qt, QML
IDEs: Arduino, VSCode
Controllers: EVCC Controller, STM32 Controller
Operating Systems: Windows, Linux, Geminix(Basics)
Version Controls: Git, Bitbucket, SVN
Protocols: I2C, SPI, UART, TCP/IP, UDP, Modbus
Development Methodologies: SDLC, Agile
Programming Skills: Multithreading, Debugging, Cross Compilation
Embedded Systems: Real-Time OS (RTOS)
Technical Skills: HW and SW Installation, Testing and Maintenance
Soft Skills: Problem Solving, Self-learning, Presentation, Adaptability, Troubleshooting

EXPERIENCE

- **Quadrant Future Tek Ltd** November 2023 - Ongoing
Software Engineer
 - Developed Driver Machine Interface (DMI) application from scratch using Qt/C++ on Linux, enabling real-time interaction between locomotive driver and LTCAS safety system.
 - Integrated DMI with LTCAS modules through TCP/IP and multicast communication, enabling real-time display of train status and safety alerts.
 - Strong knowledge of UI Designing, such as Qt Framework, security, performance, and cost optimization.
- **SaNa Software Services Pvt Ltd** March 2021 - October 2023
Embedded Developer
 - Designed and developed software for an electric vehicle (EV) charger using an EVCC controller, optimizing charging protocols and efficiency.
 - Ensured compliance with industry standards and enhanced communication between the charger and vehicle for reliable, safe operation.

PROJECTS

Railways | Quadrant Future Tek Ltd Bangalore, India

LTCAS:

- Responsible for the IOL (Input and Output Logic) module in the LTCAS railway safety system.
- Handled tag processing logic within IOL, including how the train reads RFID tags and determines the train's absolute location based on tag information.
- Managed speed-related logic under IOL, including transmission of speed data across safety levels (from PG level to SIL-4 level).

DMI:

- Designed and developed Driver Machine Interface (DMI) application from scratch using Qt and C++, enabling real-time train monitoring and safety alerts for locomotive drivers.
- Implemented user interface features by using Qt, ensuring an intuitive and efficient interaction between the driver and the system.
- Utilized [specific programming languages or frameworks] for application development.
- Collaborated with cross-functional teams to integrate DMI seamlessly with other LTCAS modules.

eWheels:

- Spearheaded the development of the eWheels module, enhancing the LTCAS platform's functionality.
- Implemented [specific features] to improve wheel-related operations and data processing.
- Employed [specific technologies or algorithms] to optimize eWheels' performance and accuracy.
- Conducted thorough testing and debugging to ensure reliability and robustness.

eRFID:

- Designed and implemented the eRFID software by using Qt with C++ language, a critical component for RFID-based identification within the LTCAS ecosystem.
- Integrated RFID reader functionalities, providing a secure and efficient means of tracking assets or individuals.
- Utilized [specific RFID protocols or libraries] to establish communication with RFID devices.
- Ensured data integrity and security measures were implemented in the eRFID module.

Electric Vehicle Chargers | Axonify Tech Systems Pvt Ltd

Hyderabad, India

Modules that we are used to Develop this Project

HMI 7 inch FHD Touch Display, EVC Controller, RFID Reader (MT626 series and RC522), Enegy Meter, Quectel EC200 Board, PT1000 Temperature Sensor, Latch Relay, LED's, RS232, RS485, Type 2 Gun (70mm)

Roles and Responsibilities:

- Generated user interfaces (UI) to the charger HMI (7 inch FHD Touch Display) by using Qt Creator.
- EVC Controller is used to turn the relay on and off. If the relay is ON, the charging will go up and if the relay is OFF, the charging will be OFF.
- For User authentication we are providing RFID as well as Application.
- Some Chargers are Configured with OCPP (Open Charge Point Protocol).
- We tested the OCPP configured chargers by using Steve, Numocity, Elocity etc., Servers.
- By Using IoT sim we established network connection between charger display and Quectel EC200 Board.
- Connected the RFID Reader (MT626 series) to HMI Display via RS232 and read the RFID serial numbers.
- I read Energy meter vaules(voltage, current and kwh) by using RS485 and Modbus.
- Performed on Pilot, Select, Elemeasure Energy Meters.
- Worked on Latch relays(On and Off)
- Knowledge on PT1000-S temperature sensor.

PERSONAL DETAILS

Date of Birth: 15:06:1997

Address: Srisailam, Nandyala Dist, Andhra Pradesh State

Languages: English, Telugu, Hindi